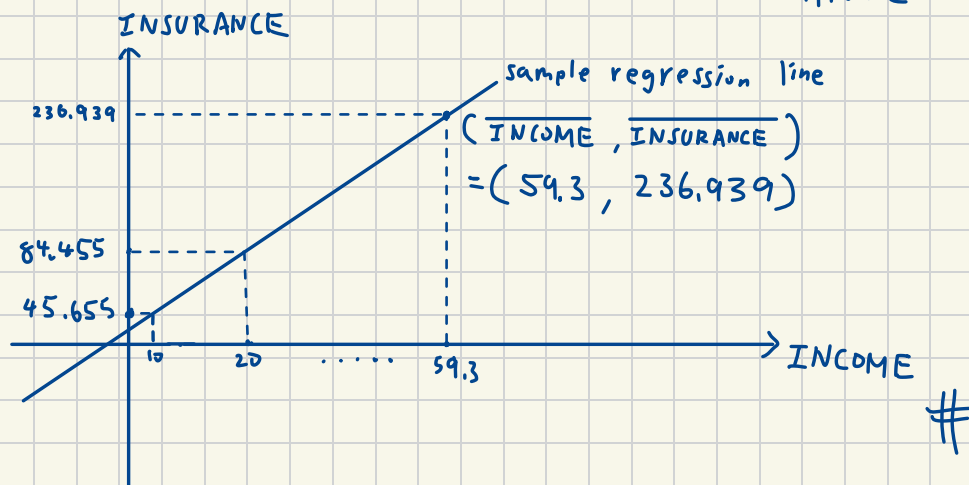


3.18

$$(a) \text{INSURANCE} = 6.855 + 3.880 \text{INCOME}$$

$$\text{INCOME} = 59.3, \text{代入上式得 } \text{INSURANCE} = 6.855 + 3.880 \times 59.3 = 236.939$$



$$(b) \hat{y} = b_1 + b_2 X \Rightarrow \hat{y} = 6.855 + 3.880X, X \text{代入}$$

$$\hat{y} = 6.855 + 3.880 \times 1 = 10.735$$

表示每家庭收入額外增加 \$1000, 平均投保金額會增加 \$10735

$$b_2 \sim N(B_2, \text{Var}(b_2))$$

$$T = \frac{b_2 - B_2}{\text{se}(b_2)} \sim t(18)$$

$$\begin{aligned} \text{The 95\% C.I. of } B_2: & [b_2 - \text{se}(b_2) \cdot t_{0.975}(18), b_2 + \text{se}(b_2) \cdot t_{0.975}(18)] \\ & = [3.880 - 0.112 \cdot 2.101, 3.880 + 0.112 \cdot 2.101] = [3.645, 4.115] \end{aligned}$$

在重複抽樣中, 當家庭收入增加 \$1000, 有 95% 的機率平均投保金額會增加 \$3645 ~ 4115 之間

(c) 當收入增加 \$100,000 ($X=100$)

$$\hat{y} = 6.855 + 3.880 \times 100 = 394.855$$

$$\hat{y} = b_1 + b_2 X$$

$$\begin{aligned}\text{Var}(\hat{y}) &= \text{Var}(b_1 + b_2 X) = \text{Var}(b_1) + X^2 \cdot \text{Var}(b_2) + 2 \cdot X \cdot \text{Cov}(b_1, b_2) \\ &= 0.383^2 + 100^2 \cdot 0.112^2 + 2 \cdot 100 \cdot (-0.746) \\ &= 30.7487\end{aligned}$$

$$\begin{aligned}\text{The } 99\% \text{ C.I. of } y &: [\hat{y} - t_{0.005}(18) \cdot \text{se}(\hat{y}), \hat{y} + t_{0.005}(18) \cdot \text{se}(\hat{y})] \\ &= [394.855 - 2.878 \cdot \sqrt{30.7487}, 394.855 + 2.878 \cdot \sqrt{30.7487}] \\ &= [378.896, 410.814] \quad \# \end{aligned}$$

(d)

$$H_0: B_2 = 5 \quad \text{vs} \quad H_1: B_2 \neq 5$$

$$\hat{y} = b_1 + b_2 X$$

$$b_2 \sim N(B_2, \text{Var}(b_2))$$

test statistic

$$T = \frac{b_2 - B_2}{\text{se}(b_2)} \sim t(18)$$

Critical value is $t_{0.025}(18) = 2.101$. The rejection region: $\{t: t > 2.101 \text{ or } t < -2.101\}$

$$\therefore T = \frac{3.88 - 5}{0.112} = -10 < -2.101 \in \text{RR} \therefore \text{reject } H_0 \text{ at } \alpha = 0.05$$

表示我們有足夠證據顯示家庭收入增加 \$1000, 平均投保金額不會增加 \$5000 #

(e)

$$H_0: \beta_2 = 1 \quad \text{vs} \quad H_1: \beta_2 > 1$$

$$\hat{y} = b_1 + b_2 X$$

$$b_2 \sim N(\beta_2, \text{Var}(b_2))$$

Test statistic

$$T = \frac{b_2 - \beta_2}{\text{se}(b_2)} \sim t(18), \text{ 拒絕域或 } RR: \left\{ t > t_{0.01}(18) = 2.552 \right\}$$

$$\therefore T = \frac{3.880 - 1}{0.112} = 25.714 > 2.552 \in RR \quad \therefore \text{reject } H_0 \text{ at } \alpha = 0.01$$

表示有足夠證據顯示 $\beta_2 > 1$ #

3.23

(a) $\text{price} = \alpha_1 + \alpha_2 \cdot \text{SOFT}^2 + e$

$$\frac{d(\text{price})}{d(\text{SOFT})} = 2 \cdot \alpha_2 \cdot \text{SOFT} = 2 \times \alpha_2 \times 20 = 40\alpha_2$$

$$H_0: 40\alpha_2 \leq 13 \quad \text{vs} \quad H_1: 40\alpha_2 > 13$$

$$\Rightarrow \alpha_2 \leq \frac{13}{40} \quad \Rightarrow \alpha_2 > \frac{13}{40}$$

$$\hat{\alpha}_2 \sim N(\alpha_2, \text{Var}(\hat{\alpha}_2))$$

test statistic

$$T = \frac{\hat{\alpha}_2 - \frac{13}{40}}{\text{se}(\hat{\alpha}_2)} \sim t(498)$$

$$\text{Rejection Region: } \left\{ t > 1.647919 \right\}$$

$$\therefore T = -26.72875 \notin RR \quad \therefore \text{Do not reject } H_0 \text{ at } \alpha = 0.05$$

表示無足夠證據顯示房子每增加2000英尺，平均房價會增加\$13000 #

```
> t_stat_a <- (a2_hat - 13/40) / (se_a2)
> p_value_a <- pt(t_stat_a, df = df, lower.tail = FALSE)
> RR_a <- qt(0.05, 498, lower.tail = FALSE)
> cat("t統計量為:", t_stat_a, "\n")
t統計量為: -26.72875
> cat("拒絕域為:t>", RR_a, "\n")
拒絕域為:t> 1.647919
> cat("p_value為:", p_value_a, "\n")
p_value為: 1
```

$$(b) \frac{d(\text{Price})}{d(\text{SOFIT})} = 2 \cdot d_2 \cdot 40 = 80 d_2$$

$$H_0: 80 d_2 \leq 13 \quad \text{vs} \quad H_1: 80 d_2 > 13$$

$$\Rightarrow d_2 \leq \frac{13}{80} \quad \Rightarrow d_2 > \frac{13}{80}$$

$$\hat{d}_2 \sim N(d_2, \text{Var}(\hat{d}_2))$$

$$\text{Test statistic}$$

$$T = \frac{\hat{d}_2 - \frac{13}{80}}{\text{se}(\hat{d}_2)} \sim t(498)$$

$$RR: \{t > 1.647919\}$$

$\therefore T = 4.189467 \in RR \therefore \text{Reject } H_0 \text{ at } \alpha = 0.05$

表示有足夠證據顯示房子每增加4000^{平方}英尺，平均房價會增加\$13000 #

(c)

```
> cat("99% Confidence Interval: [", lower_bound_y_hat, ", ", upper_bound_y_hat, "]\n")
99% Confidence Interval: [ 158.0481 , 176.6988 ]
```

表示在重複抽樣中，我們建立的所有區間中，大約有99%的機率會包含到真實 d_2

其表示在面積為2000平方英尺的房子，市場上所有這類大小房子的平均預期售價落在\$158,048至176,698之間的#機率為99% #

(d)

```
> cat("2000 sqft真實平均房價:", sample_mean_2000, "\n")
2000 sqft真實平均房價: 163.3333
```

在原始資料中，在房屋面積剛好為2,000平方英尺的樣本中，其樣本平均售價為163.3333千美元。由於此平均值落在(c)小題所估計的95%信賴區間[158.0841, 176.6988]內，因此可以結論：該樣本結果與迴歸模型的預測是相容的，代表模型能有效地反映出該特定房屋尺寸在現實樣本中的平均價格水準。

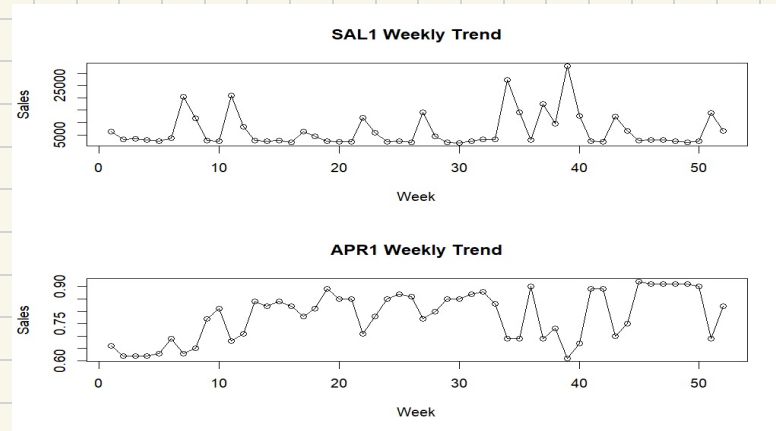
#

```
> t_stat_b <- (a2_hat - 13/80) / (se_a2)
> p_value_b <- pt(t_stat_b, df = df, lower.tail = FALSE)
> RR_b <- qt(0.05, 498, lower.tail = FALSE)
> cat("t統計量為:", t_stat_b, "\n")
t統計量為: 4.189467
> cat("拒絕域為:t>", RR_b, "\n")
拒絕域為:t> 1.647919
> cat("p_value為:", p_value_b, "\n")
p_value為: 1.65395e-05
```

3.31

(a)

```
> summary(tuna$sal1)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 1762   2485   3136   6719   8612   32820
> summary(tuna$apr1)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.6100  0.6900  0.8100  0.7825  0.8625  0.9200
> sd_sal1<-sd(tuna$sal1)
> sd_apr1<-sd(tuna$apr1)
> cat("SAL1的標準差為:",sd_sal1,"\n")
SAL1的標準差為: 6915.083
> cat("APR1的標準差為:",sd_apr1,"\n")
APR1的標準差為: 0.09803711
```

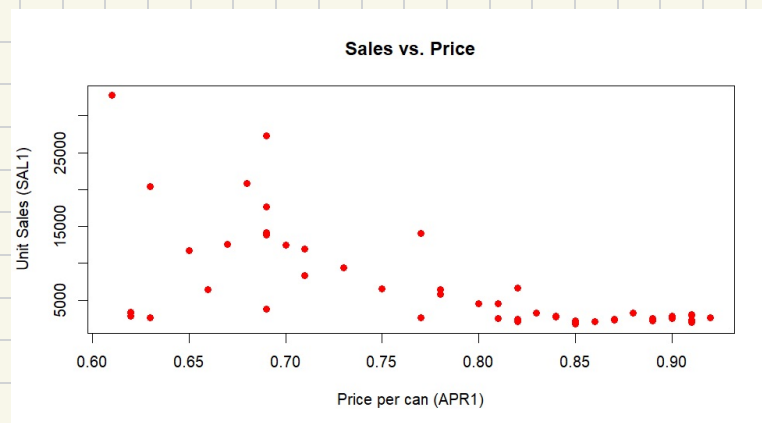


	SAL1	APR1
Sample mean	6719	0.7825
minimum	1762	0.6100
maximum	32820	0.9200
Standard deviation	6915.083	0.09803711

```
> cat(" sales的變異係數(CV):", CV_sal1, "\n")
sales的變異係數(CV): 1.029228
> cat(" price的變異係數(CV):", CV_apr1, "\n")
price的變異係數(CV): 0.125287
```

Sales 的變異係數(1.029) > price 的變異係數(0.125)
表示該魚有魚品牌的需求對每週的價格變動非常敏感 #

(b)



Sales 和 Price 呈負向關係, 此現象亦較合理, 符合我們的預期, 代表當罐頭價格上漲, 其銷售額會下降, 此符合經濟學的需求法則。 #

(c)

```
> cat("價格每增加1美分，平均銷售額變動:", b2_hat, "\n")
價格每增加1美分，平均銷售額變動: -434.4473
> cat("99% Confidence Interval:[", lower_bound, ", ", upper_bound, "]\n")
99% Confidence Interval: [ -592.2897 , -276.6049 ]
```

斜率黑點估計為 $\hat{B}_2 = -434.4473$

B_2 之 95% 信賴區間為 $[-592.2897, -276.6049]$ #

(d)

```
> cat("預期銷量 (fit):", pred_d[1, "fit"], "\n")
預期銷量 (fit): 10302.9
> cat("90%信賴區間:[", pred_d[1, "lwr"], ", ", pred_d[1, "upr"], "]\n")
90%信賴區間: [ 8624.934 , 11980.87 ]
```

當魚有魚罐頭價格為 70 cent 時，平均每週銷售量為 10302.9 罐

在價格為 70 cent 條件下，真實平均週銷量之 90% 信賴區間為 $[8624.934, 11980.87]$ 罐

(e)

```
> mean_price <- mean(tuna$price1)
> elas_hat <- b2_hat * (mean_price / mean_sal1)
> ci_b2 <- confint(model_tuna, "price1", level = 0.95)
> ci_elas <- ci_b2 * (mean_price / mean_sal1)
> cat("彈性點估計:", elas_hat, "\n")
彈性點估計: -5.059825
> cat("彈性95%信心區間:[", ci_elas[1], ", ", ci_elas[2], "]\n")
彈性95%信心區間: [ -6.898149 , -3.221501 ]
```

需求彈性估計值 = -5.06，此為 fairly elastic (彈性絕對值 > 1)

此符合經濟意義，因為魚有魚罐頭為民生消費品，市場上有許多替代品，

故若此產品一漲價，其需求會大幅減少。

#

(f)

```
> null_elas <- -3
> se_elas <- se_b2 * (mean_price / mean_sal1)
> t_stat_elas <- (elas_hat - null_elas) / se_elas
> p_value_elas <- 2 * pt(abs(t_stat_elas), df.residual(model_tuna), lower.tail = FALSE)
> t_crit_elas <- qt(0.95, df=df.residual(model_tuna))
> cat("t統計量:", t_stat_elas, "\n")
t統計量: -2.250572
> cat("p-value:", p_value_elas, "\n")
p-value: 0.02884204
> cat("拒絕域:t>", t_crit_elas, "or t<-", t_crit_elas, "\n")
拒絕域:t> 1.675905 or t<- 1.675905
```

$$H_0: \theta = -3 \quad \text{vs} \quad H_1: \theta \neq -3$$

$$\hat{\theta} \sim N(E(\hat{\theta}), \text{Var}(\hat{\theta}))$$

test statistic

$$T = \frac{\hat{\theta} - 3}{\text{se}(\hat{\theta})} \sim t(50)$$

$$RR: \{ t > 1.675905 \text{ or } t < -1.675905 \}$$

$\therefore P\text{-value} = 0.02884204 < 0.1 \therefore \text{reject } H_0 \text{ at } \alpha = 0.1$

表示有足夠證據顯示價格需求彈性不等於-3 #